

CSE 479

Adaptive Cruise Control Project

Fall 2025

Objective

In this project, you will design and implement an adaptive cruise control system using the EK-TM4C123GXL Tiva C microcontroller on a 4-wheel car platform.

Description

Build a small-scale adaptive cruise control car that can automatically adjust its speed to maintain a safe following distance from objects or vehicles ahead. At a high level, you will initialize the hardware, read distance information from a sensor, control the motors with PWM, and implement a basic control loop across a few key subroutines (such as *system_init*, *ultrasonic_read*, *motor_control*, and *acc_control_loop*).

Extra Credit

- If you submit a working program prior to the ACC Light Safety Net Lecture on Friday, November 21, 2025, you will be eligible for ten points extra credit.
- If you submit a working program after the ACC Light Safety Net Lecture on Friday, November 21, 2025, but before the ACC Safety Net Lecture on Wednesday, November 26, 2025 you will be eligible for five points extra credit.

Documentation

Document only your *system_init*, *ultrasonic_read*, *motor_control*, and *acc_control_loop* routines.

Submissions

Your source code (C and assembly) must be submitted online using the submit command (*submit_cse479 acc_wrapper.c acc.s acc_library.s acc_documentation.pdf*) on turing.cse.buffalo.edu before 1:00 PM on Friday, December 5, 2025. Note if you did not use a library file or you reduced the number of files from those given in the list, that is fine. Just submit the ones you used. What is important is to keep the names of the files the same as in the above list so we can easily locate them in the submit directory.